

Mohit Vaid

The George Washington University | (202)-7131164 | m.vaid@gwu.edu | [LinkedIn](#) | github.com/mvaid1

RESEARCH INTEREST

Secure Multiparty Computation with current focus on Distributed Point Functions, Private Set Intersections; Functional Encryption ; Broader interests include Differential Privacy, Content Moderation in E2E encrypted messaging services; Post Quantum Cryptography involving SQISign signature schemes.

EDUCATION

The George Washington University

Washington, DC

Doctor of Philosophy, Cryptography

May 2029

- Relevant Coursework : Design and Analysis of Algorithms, Introduction to Cryptography, Differential Privacy, Advanced Software Paradigms

The Johns Hopkins University

Baltimore, MD

Master of Science

May 2024

- Relevant Coursework : Practical Cryptography, Security and Privacy in Computing, Intrusion Detection, Intro to Algorithms

Indian Institute of Technology Kanpur

Kanpur, India

Master of Technology

July 2022

- Relevant Coursework : Numerical Methods, Probability and Statistics

National Institute of Technology Kurukshetra

Kurukshetra, India

Bachelor of Technology

July 2016

- Relevant Coursework: Engineering Mathematics, Computer Programming

PRESENT PROJECTS

Updatable Private Set Intersection

Oct 2025 - till date

- Exploring the existing 2-party UPSI protocol and trying to develop a multi party version of the same with succinct communication and computation complexity.

Functional Encryption

June 2025- till date

- Developing a Functional Encryption Scheme for some constraint function class to work in multi-client setting under the guidance of Prof Arkady Yeruchimovich

Distributed Point Function

August 2024 - till date

- Developing protocol for multiparty distributed point function for dishonest majority under the Supervision of Prof Arkady Yeruchimovich at The George Washington University

PAST PROJECTS

Real Time Proctoring Application

January 2025 - May 2025

- Developed a Python-based proctoring system leveraging MediaPipe Face Mesh and OpenCV for real-time blink rate and head movement analysis to detect cheating behavior during online interviews, featuring adjustable detection thresholds and automated session logging. [link](#)

Blockchain Based Ransomware Proof-of-Concept

January 2024-May 2024

- Leveraged Ethereum smart contracts to build an automated escrow service for fair exchange mechanism whilst exploring technical feasibility and security implications under the supervision of Prof Mathew Green at the Johns Hopkins University. [link](#)

Padding Oracle attack on E2E Messaging Client

January 2024- May 2024

- Developed with one other an End to End Messaging Client based on ECDH, ChaCha20 and ECDSA, and conducted padding oracle attacks to study the security risks for the client, under Prof Mathew Green at Johns Hopkins University. [link](#)

WORKING EXPERIENCE

Dunnhumby India Limited

Gurugram, India

Junior Developer

July 2022 -July 2023

- Developed UI and API based upon react and C# respectively for displaying the real time Jobs and also the options to cancel, delete or modify Jobs

HSCC India Limited

Noida, India

Project Management

July 2016 - July 2019

TEACHING EXPERIENCE

Numerical Methods , IIT Kanpur India

August 2021-December 2021

Atmospheric Physics and Chemistry, IIT Kanpur India

January 2022 - May 2022

CONFERENCES ATTENDED

PETS 2025

Washington DC, USA

Indocrypt 2024

Chennai, India